SAIFA QUANT EDGE 1.0 FINAL CHALLENGE

SERENDIB UNIVERSITY ENDOWMENT FUND

Tracking Code : SAIFA-2026-11FCECCB

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FINAL RECOMMENDATION

Adopt the Best Risk Parity portfolio as the strategic model allocation:
65.45% AGG
17.79% GLD
16.76% XLK.
The recommendation prioritizes capital preservation, validated downside-risk measurement and stress resilience over return maximization.

Mandate	Capital base	Evaluation period	Recommended portfolio
Conservative institutional endowment	USD 10 million	Development through 31 Dec 2021; OOS from 3 Jan 2022	Best Risk Parity

Scope. Prepared by Quant_Colombo using the prescribed ETF universe, Yahoo Finance adjusted prices and FRED 3-month US Treasury data. All portfolio weights are frozen before out-of-sample evaluation.

Purpose. This memo is written for an investment committee. Technical detail is included where it changes the decision; implementation choices are explained in plain language so the recommendation can be defended.

Executive Summary

Mandate. The client is a USD 10 million conservative university endowment whose stated priority is to preserve and grow capital while remaining resilient in adverse markets. The project therefore treats downside control as a decision constraint, not as an afterthought. A portfolio with the highest realized return is not automatically the most suitable portfolio.

Asset	Weight	Role
AGG	65.45%	Defensive fixed-income anchor
GLD	17.79%	Real-asset / crisis diversifier
XLK	16.76%	Growth and upside participation

Out-of-sample evidence. From 3 January 2022 to 19 October 2026, the recommended portfolio produced a 6.51% annualized return with 8.19% volatility, a Sharpe ratio of 0.30, and a maximum drawdown of -17.11%.

Risk-model credibility. At 99% confidence, Historical VaR recorded 15 exceptions over 1,203 forecasts (1.25% observed versus 1.00% expected). Kupiec p=0.407, independence p=0.178, and conditional-coverage p=0.286; all pass at 5%. The latest Basel zone is Green.

WHY RISK PARITY?

Equal Weight earned more out of sample, but ranked last under the combined stress framework. Minimum ES ranked first under stress, but concentrated about 93% in AGG and produced a negative out-of-sample Sharpe ratio. Risk Parity is the only shortlisted portfolio that passes all four internal decision screens: validated 99% Historical VaR, positive OOS Sharpe, top-3 stress rank, and concentration below the 80% comparison screen.

Decision rule. The recommendation should be reviewed if risk-adjusted performance deteriorates persistently, VaR validation begins to fail, concentration increases materially, or stress/tail-loss estimates rise sharply.

Client Mandate, Data and Governance

Client objective. The competition brief describes a long-term endowment supporting scholarships, research, faculty development and strategic projects. The client emphasizes capital preservation, risk management, income generation and sustainable real growth, with particular concern for large losses during stressed markets. No numerical spending rate or required return is specified, so this analysis does not invent one.

WHY WE DID NOT SET AN ARBITRARY RETURN TARGET

A required return should follow from the endowment's actual spending policy, inflation objective, fees and liabilities. Because those inputs were not supplied, the portfolio is evaluated as a risk-controlled strategic allocation rather than against a fabricated hurdle rate.

Data sources. The permitted universe contains SPY, EFA, EEM, AGG, GLD, XLE and XLK. Prices are downloaded with yfinance using auto_adjust=True, so the saved Close series is adjusted for dividends and splits. The analysis uses the common seven-asset calendar from 18 November 2004 to 19 October 2026.

Governance check	Result
Common synchronized observations	5,513
Internal missing values	0
Duplicate dates	0
Non-positive prices	0
Potential stale-price runs	0
Large-move flags retained	3

Cleaning discipline. Rows with any unavailable asset price are excluded from the synchronized portfolio sample rather than forward-filled. Large daily moves are flagged, reviewed and retained unless confirmed as data errors. This avoids silently deleting genuine crisis information that is important for tail-risk analysis.

WHY SIMPLE RETURNS AND LOG RETURNS ARE BOTH KEPT

Simple returns aggregate naturally in a weighted portfolio, so they are used for portfolio performance and risk calculations. Log returns are retained for statistical diagnostics and dependence analysis because they are convenient for time-series work. The report states explicitly where each is used.

Statistical Diagnostics and Institutional KPIs

Diagnostics. All seven return series reject normality under the Jarque-Bera test at the 5% level and exhibit substantial excess kurtosis. At the same time, the Augmented Dickey-Fuller test rejects a unit root for every log-return series, so returns are treated as stationary for subsequent modelling.

WHY THIS MATTERS FOR RISK MODELLING

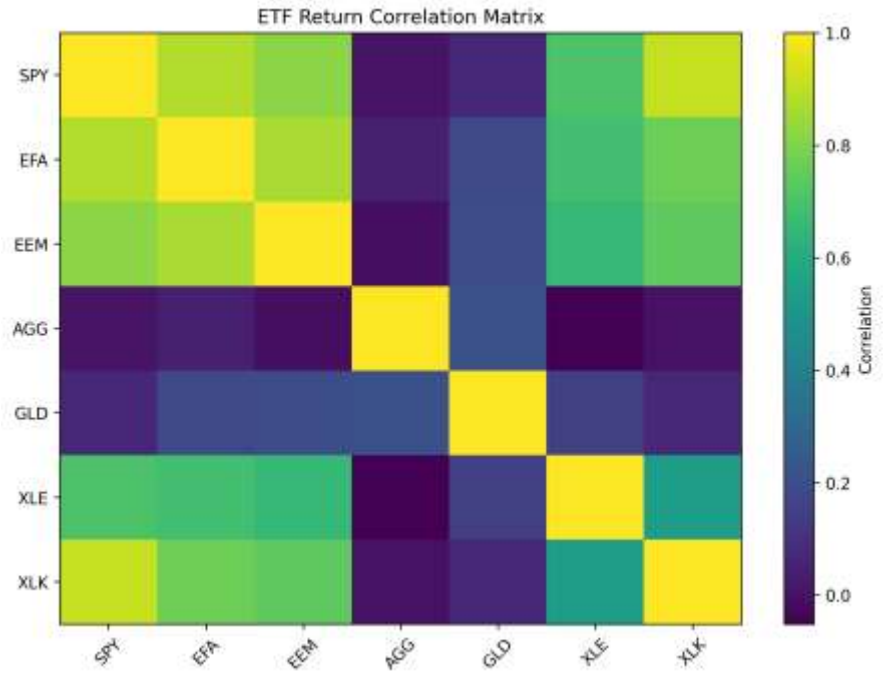
If returns were close to normal, a parametric Normal VaR model might be sufficient. The observed heavy tails mean extreme losses occur more often than a normal model suggests. That directly motivates Historical VaR, Expected Shortfall and EVT as complementary downside-risk tools.

CAPM/KPI framework. SPY is used as the market benchmark and the FRED 3-month US Treasury yield (DGS3MO) as the risk-free-rate input. CAPM and institutional KPIs are calculated on development data only.

Asset	Ann. return	Volatility	Sharpe	Sortino	Max DD
SPY	11.9%	19.2%	0.56	0.78	-55.2%
AGG	3.9%	4.9%	0.56	0.76	-12.8%
GLD	9.5%	18.0%	0.46	0.65	-45.6%
XLK	16.1%	21.5%	0.69	0.99	-53.0%

Investment interpretation. XLK provides the strongest development-sample return and risk-adjusted performance, while AGG is the defensive anchor with low volatility and a shallow development-sample drawdown. GLD adds a different return driver and crisis-diversification potential. These characteristics help explain why the final Risk Parity subset is AGG-GLD-XLK.

Figure 1. Cross-asset correlation structure. Source: Notebook 02.



Portfolio Construction and Candidate Selection

Construction rules. Portfolio construction uses development data only through 31 December 2021. Expected returns are annualized historical arithmetic means. Covariance is estimated using Ledoit-Wolf shrinkage. All optimized portfolios are long-only, fully invested, unlevered, include between two and seven assets, and require at least 5% in every included position.

WHY LEDOIT-WOLF COVARIANCE

Sample covariance can be noisy when many correlations and volatilities must be estimated from finite data. Ledoit-Wolf shrinkage pulls unstable covariance estimates toward a more structured target, reducing estimation error and making optimizer outputs less sensitive to sample noise.

WHY THE 5% MINIMUM POSITION EXISTS

Without a minimum, an optimizer can "include" an asset with an economically trivial weight. The 5% floor makes subset comparisons meaningful. It is a modelling assumption, not a client policy, and is disclosed as such.

Candidate set. Five construction methods are compared: Equal Weight, Global Minimum Variance (GMVP), Maximum Sharpe, Risk Parity and Minimum Historical Expected Shortfall (95%). Every eligible subset is enumerated, producing 600 candidate portfolios (120 subsets x 5 methods) with zero optimizer failures.

Shortlisted portfolio	Assets	Method
Equal Weight Benchmark	SPY, EFA, EEM, AGG, GLD, XLE, XLK	Equal Weight
Best GMVP	SPY, AGG	GMVP
Best Maximum Sharpe	AGG, GLD, XLK	Maximum Sharpe
Best Risk Parity	AGG, GLD, XLK	Risk Parity
Best Minimum ES	SPY, AGG	Minimum ES

WHY RISK PARITY IS NOT "EQUAL WEIGHT"

Risk Parity equalizes risk contribution, not capital weight. Lower-volatility AGG therefore needs a larger dollar allocation to contribute roughly the same portfolio risk as GLD and XLK. This distinction is central to the final recommendation.

Out-of-Sample Performance and the Core Trade-Off

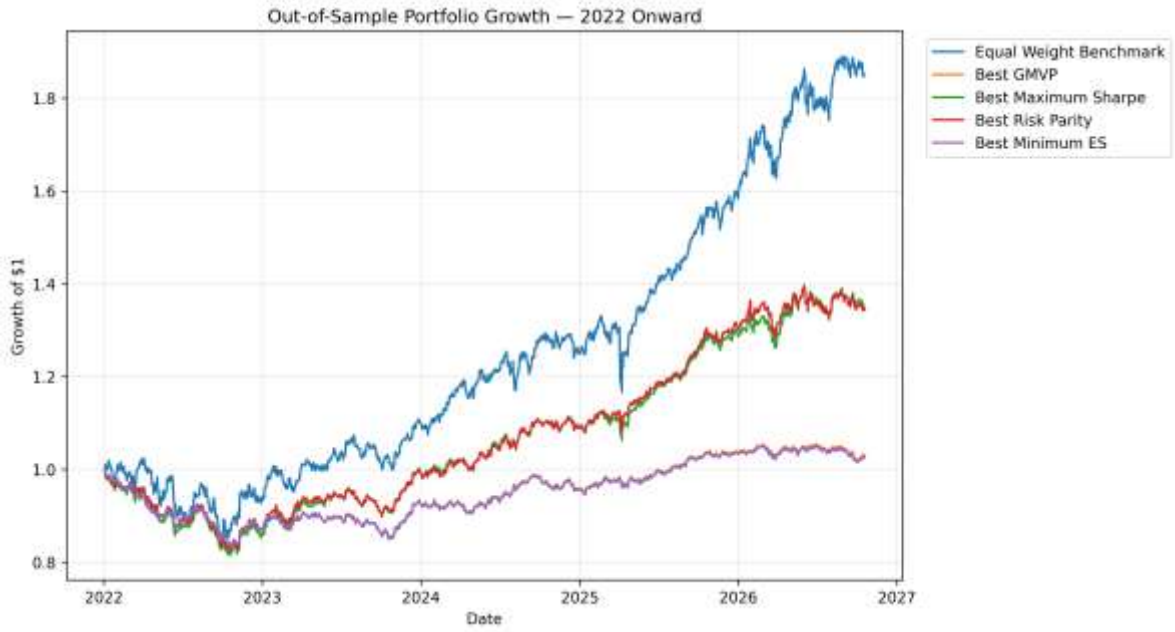


Figure 2. Frozen portfolio growth from 3 January 2022 onward. Source: Notebook 06.

Portfolio	Return	Volatility	Sharpe	Max DD
Equal Weight Benchmark	13.76%	13.30%	0.73	-17.37%
Best GMVP	0.79%	6.19%	-0.53	-16.44%
Best Maximum Sharpe	6.63%	8.62%	0.30	-18.10%
Best Risk Parity	6.51%	8.19%	0.30	-17.11%
Best Minimum ES	0.72%	6.18%	-0.54	-16.41%

WHY NOT EQUAL WEIGHT?

Equal Weight is the strongest realized performer, with a 13.76% annualized return and 0.73 Sharpe ratio. However, it ranks fifth under the combined stress framework. For a client whose explicit priority is avoiding large losses, realized return cannot be the only selection rule.

WHY NOT GMVP OR MINIMUM ES?

Both defensive portfolios reduce volatility and drawdown, but each allocates more than 92% to AGG and produces a negative out-of-sample Sharpe ratio. They show the cost of optimizing one downside dimension too aggressively: strong protection can become return-constrained and concentrated.

Trade-off conclusion. Maximum Sharpe remains a credible return-oriented alternative, but its stress rank is fourth. Risk Parity preserves a positive OOS Sharpe ratio, ranks second under stress and avoids the extreme bond concentration of the most defensive candidates.

VaR, Expected Shortfall and Extreme Value Theory

Model	99% VaR	99% ES
Historical	0.939%	1.458%
Parametric Normal	0.848%	0.976%
Monte Carlo Normal	0.849%	0.976%
EVT	0.958%	1.436%

VaR vs ES. VaR is the loss threshold associated with a selected confidence level. Expected Shortfall answers a different question: once VaR is exceeded, how large is the average loss? For a capital-preservation mandate, ES is essential because it measures severity beyond the cutoff.

WHY COMPARE THREE VaR METHODS

Historical VaR uses the empirical return distribution, Parametric VaR imposes a Normal distribution, and Monte Carlo samples from an estimated multivariate Normal model. Comparing them reveals model risk. At 99%, Historical VaR for Risk Parity is about 0.91 percentage points higher than Parametric VaR, evidence that the normal assumption matters in the tail.

WHY EVT AND WHY A 95TH-PERCENTILE THRESHOLD

Peaks-over-Threshold EVT focuses specifically on extreme losses. Thresholds of 90%, 92.5%, 95% and 97.5% were tested. The 95th percentile is retained as a practical compromise: it isolates the extreme tail while preserving about 216 exceedances, compared with only about 108 at 97.5%. Nearby thresholds are kept as sensitivity checks instead of being ignored.

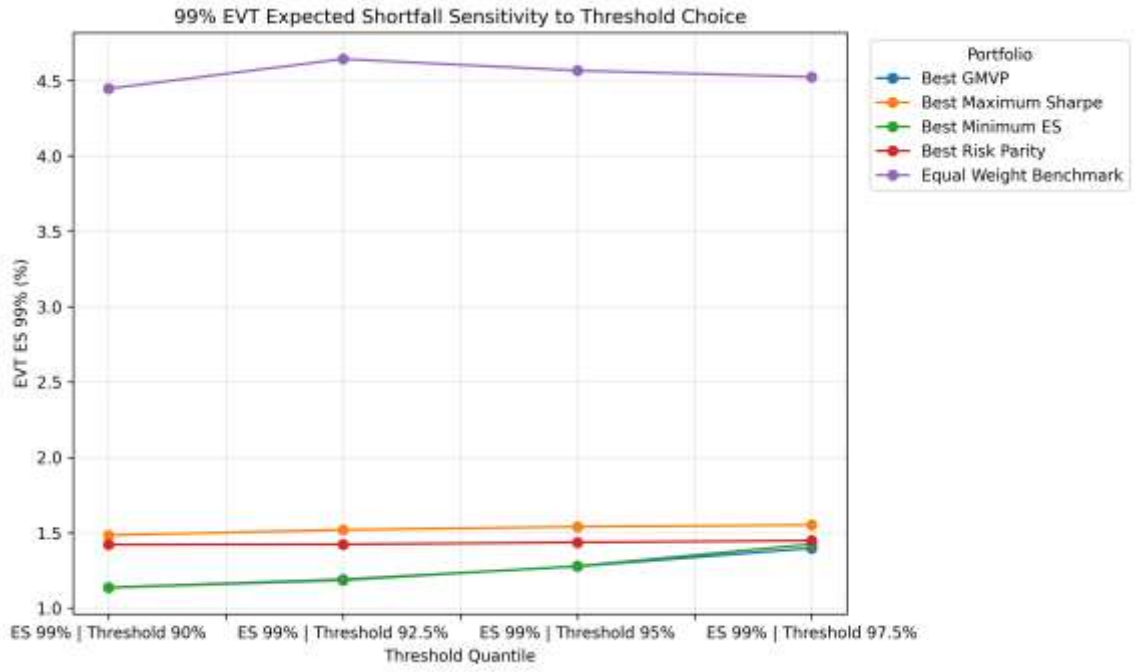


Figure 3. EVT 99% Expected Shortfall threshold sensitivity. Source: Notebook 05.

Out-of-Sample VaR Backtesting

Design. Portfolio weights are frozen before the out-of-sample period. Historical and Parametric Normal VaR forecasts use a rolling 500-trading-day estimation window, with each forecast based only on information available before that date.

WHY A ROLLING 500-DAY WINDOW

A rolling window lets the risk estimate adapt as market conditions change while still retaining roughly two years of daily observations. It is long enough to estimate a 99% quantile more credibly than a very short window, but short enough to respond to regime changes. Most importantly, the procedure is strictly backward-looking at each forecast date.

Risk Parity validation	Result
Forecasts	1,203
99% Historical VaR exceptions	15
Observed exception rate	1.25%
Expected exception rate	1.00%
Kupiec p-value	0.407
Independence p-value	0.178
Conditional coverage p-value	0.286
Latest Basel zone	Green

Result. Historical VaR passes unconditional coverage, independence and conditional coverage for all five shortlisted portfolios. Parametric Normal VaR fails unconditional and conditional coverage across the portfolio set and produces too many 99% exceedances.

WHY HISTORICAL VaR BECOMES THE MONITORING MODEL

The earlier normality tests already suggested that a Normal model could understate tail risk. Backtesting confirms that concern out of sample. Historical VaR is therefore preferred for regular monitoring, while ES, EVT and copula-based estimates remain complementary tail-severity checks.

Stress Testing and Scenario Analysis

Historical stress. Historical replays cover the Global Financial Crisis (2008), the March 2020 COVID crash and the 2022 inflation/rate shock. The 2008 and 2020 windows fall inside the development sample and are therefore stress replays, not independent validation. The 2022 window is genuinely out of sample.

Hypothetical stress. The hypothetical set applies deterministic shocks for a severe equity crash, inflation/rate shock, technology-sector crash and emerging-market crisis. These are sensitivity tests, not forecasts and not probability-weighted scenarios.

WHY STRESS TESTS ARE NEEDED IN ADDITION TO VaR

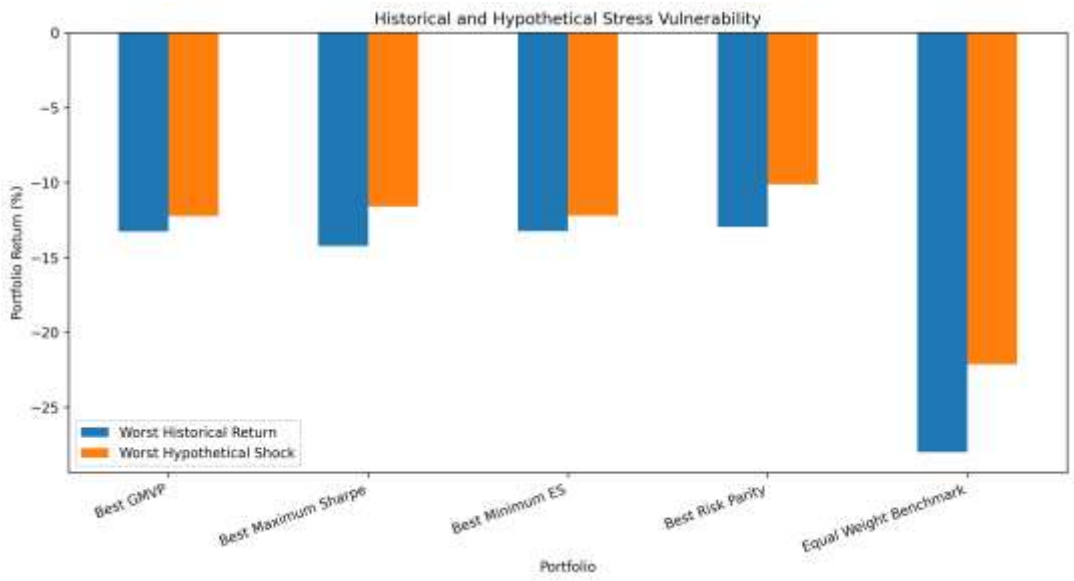
VaR and ES summarize a distribution; stress tests ask what happens under named market states that may be rare, structurally different, or poorly represented by recent data. For an endowment, this translates statistical risk into capital outcomes that an investment committee can discuss directly.

Risk Parity historical replay	Cumulative return	Max drawdown
GFC 2008	-0.15%	-12.45%
COVID crash 2020	-6.63%	-10.50%
2022 rate shock	-12.97%	-17.11%

STRESS DECISION

Minimum ES ranks first, Risk Parity second, GMVP third, Maximum Sharpe fourth and Equal Weight fifth. Stress rank is a transparent decision aid, not a formal optimization objective; it is combined with OOS performance, concentration and risk-budgeting evidence in the final selection.

Figure 4. Combined stress-resilience comparison. Lower rank is better. Source: Notebook 07.



Risk Budgeting and Final Portfolio Selection

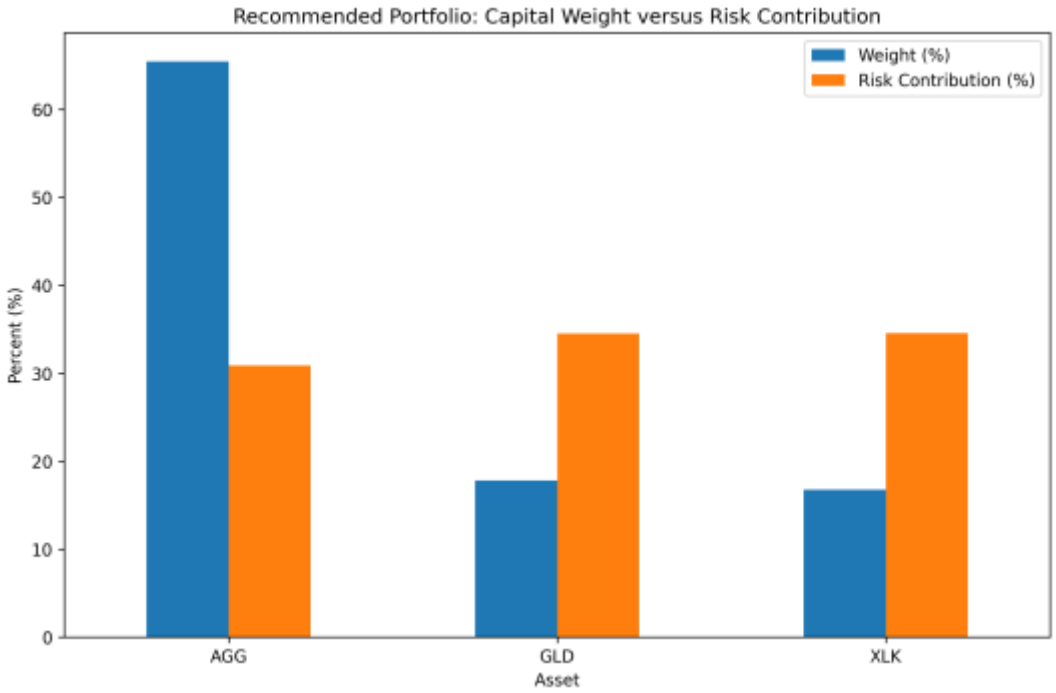


Figure 5. Capital weights versus volatility-risk contributions for the recommended portfolio. Source: Notebook 08.

Asset	Capital weight	Component VaR share	Component ES share
AGG	65.45%	30.88%	32.66%
GLD	17.79%	34.54%	31.23%
XLK	16.76%	34.58%	36.11%

Risk-balance result. The allocation is not equal by capital, but the volatility-risk contributions are approximately 30.9% AGG, 34.5% GLD and 34.6% XLK. Tail-risk budgeting tells a similar story: 99% component ES shares are about 32.7%, 31.2% and 36.1%.

WHY THIS IS IMPORTANT

A portfolio can look diversified by market value while one position dominates actual downside risk. Marginal and component VaR/ES show where risk is coming from. In Risk Parity, the three holdings contribute far more evenly to risk than their capital weights suggest.

Institutional requirement	Assessment	Evidence
Capital preservation	Strong	8.19% OOS volatility; lower crisis exposure than Equal Weight
VaR credibility	Pass	99% Historical VaR passes coverage and independence tests
Stress resilience	Strong	Overall stress rank 2 of 5
Risk diversification	Strong	Volatility contributions about 31% / 35% / 35%
Risk-adjusted return	Positive	OOS Sharpe ratio about 0.30

Tail Dependence and Model-Risk Extension

Purpose. Covariance and ordinary correlation describe average co-movement, but extreme events can create stronger joint dependence. The optional copula extension therefore compares Normal Monte Carlo, Gaussian copula, and Student-t copula risk for the frozen recommended portfolio.

WHY A STUDENT-t COPULA

The Gaussian copula has zero asymptotic tail dependence when correlation is below one. The Student-t copula allows extreme moves to occur jointly with positive tail dependence. It therefore tests whether conventional Gaussian dependence assumptions understate severe portfolio losses.

Model	VaR 95	VaR 99	ES 95	ES 99
Normal Monte Carlo	0.590%	0.849%	0.751%	0.976%
Gaussian Copula	0.541%	0.935%	0.802%	1.285%
Student-t Copula	0.523%	0.963%	0.832%	1.443%

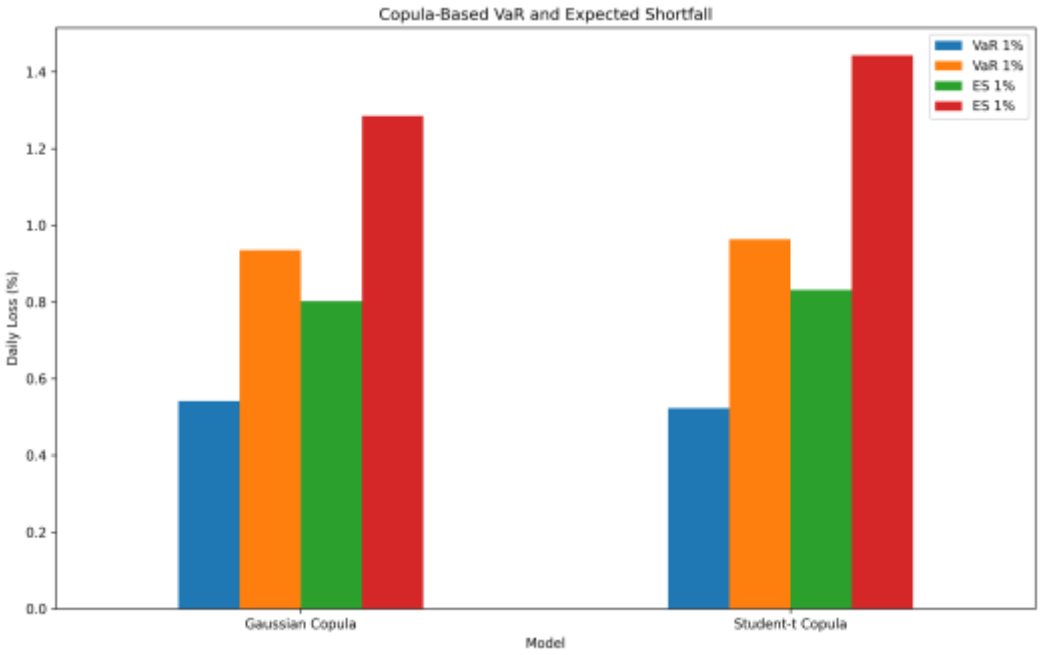


Figure 6. Normal Monte Carlo versus Gaussian and Student-t copula risk. Source: Notebook 09.

Result. Relative to Normal Monte Carlo, the Student-t copula increases 99% VaR by about 13.5% and 99% ES by about 47.9%. Relative to the Gaussian copula, the uplift is about 3.0% for 99% VaR and 12.3% for 99% ES. The deeper tail is therefore more sensitive to dependence assumptions than moderate quantiles.

INVESTMENT IMPLICATION

The copula extension does not overturn the allocation. Instead, it strengthens the case for monitoring extreme risk with multiple models. For this endowment, the severity of losses beyond VaR matters at least as much as the VaR threshold itself.

Mandate Translation and Decision Framework

The recommendation is not based on a single optimizer output. It translates the endowment mandate into a set of investment decisions and then tests whether the proposed allocation remains defensible across performance, concentration, downside risk and stress evidence.

Manager responsibility	Our position and defence
Investment horizon	Long-term / perpetual-style strategic horizon. A university endowment must support recurring academic needs across market cycles, so short-term return maximization is inconsistent with the mandate.
Return objective	Preserve purchasing power and achieve sustainable long-term growth while controlling downside risk. No numerical hurdle is imposed because the brief does not provide a spending rule, inflation target, fee budget or liability schedule.
Asset allocation	65.45% AGG, 17.79% GLD and 16.76% XLK. AGG provides stability, GLD provides a distinct real-asset / crisis driver, and XLK preserves participation in long-run growth.
Risk constraints	Long-only, fully invested, no leverage; 2-7 included assets; 5% minimum position for included assets; frozen weights before OOS evaluation. The 5% floor and 80% concentration screen are modelling/comparison rules, not client policy.
Suitability test	The portfolio must show positive risk-adjusted OOS performance, validated downside measurement, stress resilience and acceptable concentration. Risk Parity passes all four internal decision screens.
Return-risk trade-off	Equal Weight earns more but ranks last under stress. GMVP and Minimum ES protect downside but become >92% AGG and have negative OOS Sharpe ratios. Risk Parity accepts a lower return than Equal Weight in exchange for materially stronger resilience and diversification.

WHY THIS ALLOCATION IS SUITABLE FOR A CONSERVATIVE INSTITUTIONAL INVESTOR

It does not attempt to eliminate risk. Instead, it allocates risk deliberately: a large low-volatility bond position stabilizes capital, gold diversifies crisis and inflation exposures, and technology retains growth potential. The resulting volatility-risk contributions are roughly 31% / 35% / 35%, far more balanced than the capital weights.

Investment Committee Risk Questions

1. WHAT CAN GO WRONG?	2. HOW BAD COULD IT GET?	3. HOW CONFIDENT ARE WE?
Inflation and rising real yields can hurt bonds; technology can suffer valuation compression; gold may fail to hedge every crisis; correlations can rise in stress; return, volatility and tail relationships can shift by regime; model assumptions can break; transaction costs, liquidity and market impact are not explicitly modelled.	At 99% confidence, Historical VaR is 0.939% (~USD 93,928 for USD 10m) Historical ES is 1.458% (~USD 145,829). The worst historical replay is 2022 at -12.97% (~USD 1.30m). The worst designed hypothetical shock is about -10.12% (~USD 1.01m).	Historical VaR produced 15 exceptions over 1,203 OOS forecasts (1.25% observed vs 1.00% expected). Kupiec p=0.407, independence p=0.178 and conditional coverage p=0.286; Basel is Green. Passing tests supports credibility over the sample, but does not guarantee future accuracy.

Confidence is built from model comparison, not from one number. Historical, Parametric and Monte Carlo VaR are compared; ES measures severity beyond VaR; EVT focuses on extreme observations; and the Student-t copula tests whether joint tail dependence makes losses worse. The fact that Student-t 99% ES is about 47.9% above Normal Monte Carlo ES is treated as model-risk evidence, not ignored.

DIVERSIFICATION AND RESILIENCE EVIDENCE

The recommended portfolio combines three different economic roles and balances risk contribution even though capital weights are unequal. It ranks #2 of 5 in the combined stress framework, while the highest-return Equal Weight portfolio ranks #5. This is the central client trade-off: give up some upside to improve the quality and defensibility of downside protection.

Additional macro and market risks. The portfolio remains exposed to inflation/rate shocks, changing stock-bond correlations, technology-sector valuation risk, changes in gold's diversification behaviour, and shifts in market regime. These risks are monitored through stress tests, concentration/risk-contribution checks, VaR exceptions and deep-tail models rather than assumed away.

GOVERNANCE IMPLICATION

The strategic allocation should be reviewed if OOS risk-adjusted performance deteriorates persistently, VaR validation begins to fail, concentration or component risk rises materially, or stress / tail-loss estimates worsen. Rebalancing is a governance response to changing evidence, not a reaction to short-term noise.

Assumptions, Limitations and Governance

- Historical mean returns are used as baseline expected-return inputs; they are not forecasts.
- Historical relationships in volatility, correlation and tail dependence can change across regimes.
- Historical VaR assumes the past contains useful information about future downside risk; Parametric and Normal Monte Carlo models impose normality assumptions that are challenged by the data.
- EVT results depend on threshold selection. The 95th percentile is a justified baseline, not a uniquely correct threshold.
- Copula results are model-dependent. The Student-t degrees of freedom are selected by grid-based pseudo-likelihood rather than full joint maximum likelihood.
- Empirical copula marginals cannot generate asset-specific outcomes beyond historical support; EVT remains the dedicated extrapolation tool for extreme losses.
- Transaction costs, taxes, liquidity frictions, market impact and explicit rebalancing costs are not modelled.
- The endowment's exact spending rate, inflation target, liability schedule and formal required return were not supplied. Therefore the 6.5% OOS return is evidence, not a guaranteed adequacy test.
- The 2008 and 2020 crises are development-sample replays; only 2022 is an independent stress period. The COVID window is only 24 observations, so 99% tail statistics there are sample-sensitive.
- PASS/WATCH screens (including the 80% concentration screen) are internal comparison heuristics, not client-imposed policy limits.

GOVERNANCE RECOMMENDATION

Revalidate the allocation periodically using updated OOS performance, concentration, risk contributions, VaR exceptions and stress tests. Review or rebalance if risk-adjusted performance deteriorates persistently, model validation fails repeatedly, a holding or risk contribution becomes materially concentrated, or deep-tail estimates rise sharply.

Conclusion. The final model portfolio is 65.45% AGG, 17.79% GLD and 16.76% XLK. It is recommended because it best matches the client's stated priority of preserving capital while maintaining positive return potential and defensible, tested downside risk.

AI-use disclosure. AI tools were used for learning support, debugging, code review, Streamlit/dashboard development, documentation and wording refinement. The team reviewed the final methodology, code, calculations, assumptions, outputs and recommendation and remains responsible for reproducing and defending the submission.

Sources

- Yahoo Finance via yfinance: adjusted ETF price data for SPY, EFA, EEM, AGG, GLD, XLE and XLK.
- Federal Reserve Economic Data (FRED): DGS3MO, 3-month U.S. Treasury yield.
- Quant_Colombo Notebooks 01-09 and saved outputs in outputs/tables and outputs/figures.